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Chapter 5

Business Statistics

Math for Business and Finance: an Algebraic Approach, 3rd Edition
Jeffrey Slater

Learning Unit Objectives

LU 5-1: Frequency Distributions and Graphs

1. Prepare a frequency distribution.
2. Prepare bar, line, and circle graphs.
3. Calculate price relatives and cost comparisons.

LU 5-2: Mean, Median, and Mode

1. Define and calculate the mean.
2. Explain and calculate a weighted mean.
3. Define and calculate the median.
4. Define and identify the mode.

LU 5-3: Measures of Dispersion

1. Explain and calculate the range.
2. Define and calculate the standard deviation.
3. **Define normal distribution and** estimate percentage of data by using standard deviations.

Frequency Distribution

Frequency Distribution - A way of collecting and organizing raw data. Shows the number of data observations that fall into specific intervals.

Computer costs

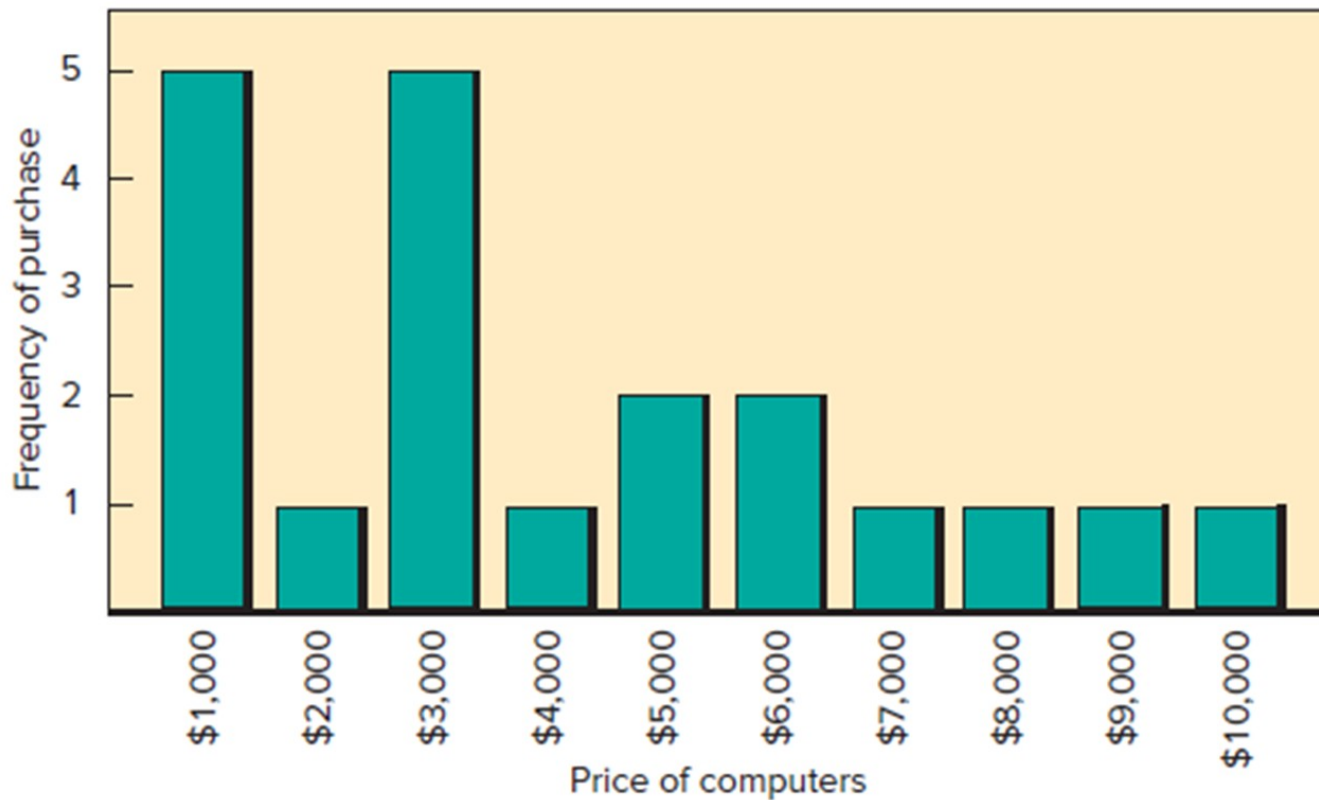
1000	7000	4000	1000	5000	1000	3000
5000	2000	3000	3000	3000	8000	9000
3000	6000	6000	1000	10000	1000	

Price of Computer	Tally	Frequency
\$1,000		5
2,000	I	1
3,000		5
4,000	I	1
5,000	II	2
6,000	II	2
7,000	I	1
8,000	I	1
9,000	I	1
10,000	I	1

Frequency distribution table

Bar Graph ¹

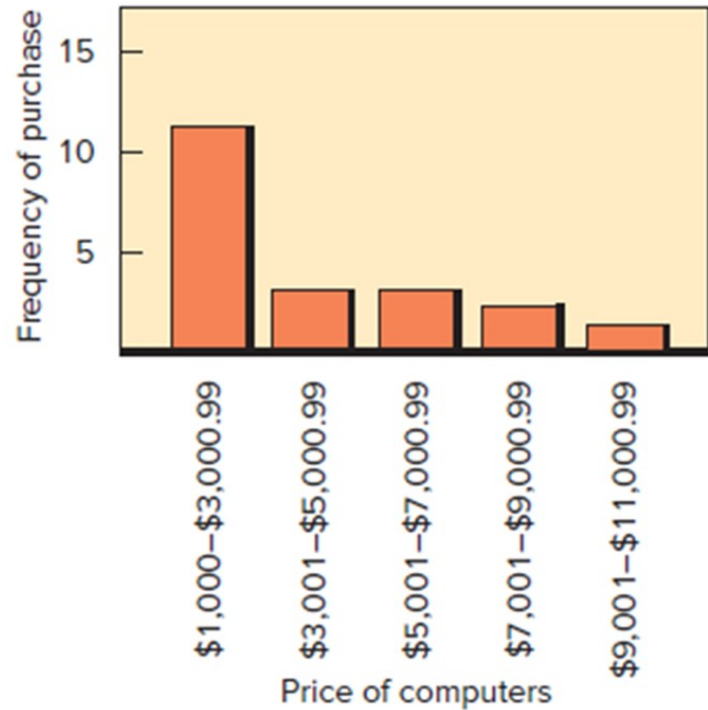
Help readers see changes that have occurred over a period of time.



[Access the text alternative for these images](#)

Bar Graph ²

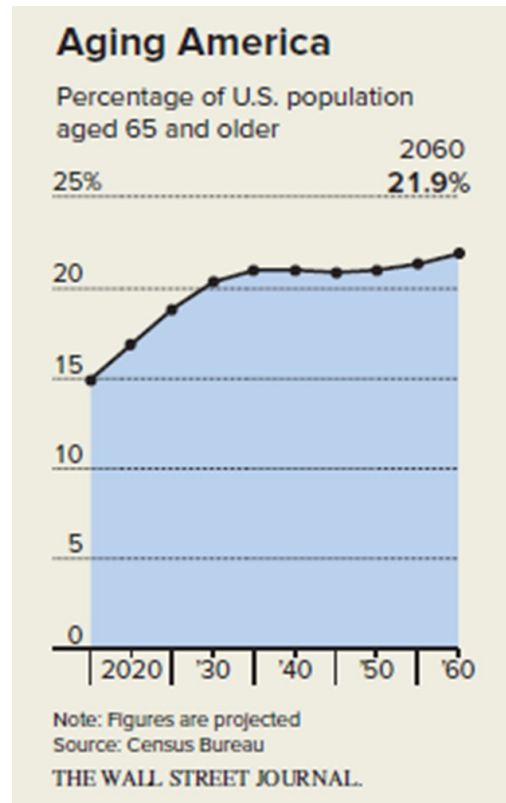
Class	Frequency
\$1,000 - \$3,000.99	11
3.001- 5.000.99	3
5.001- 7.000.99	3
7.001- 9.000.99	2
9.001- 11.000.99	1



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Line Graph

Shows a trend over a period of time.

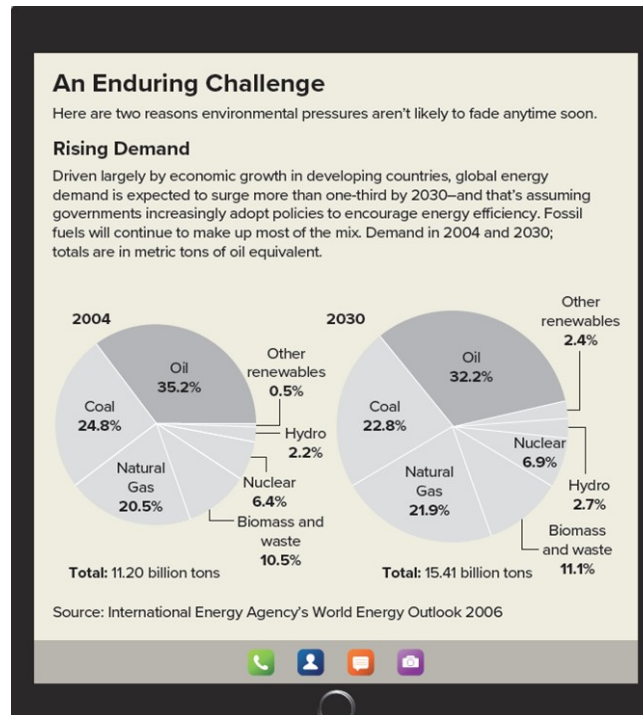


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Circle Graph (pie chart)

Helpful in showing the relationship between parts to a whole.

The entire circle represents 100%, or 360°.



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Index Numbers

Frequently used to make comparisons of a current price relative to a given year.

Also used to estimate current prices at various geographic locations.

$$\text{Price relative} = \frac{\text{Current price}}{\text{Base year's price}} \times 100$$

Example:

A calculator may cost \$9 today relative to a cost of \$75 some 30 years ago. What is the relative price?

$$\frac{\$9}{\$75} \times 100 = .12 = \mathbf{12\%}$$

Terminology

Mean – An arithmetic average found by adding all of the data and dividing by the number of values.

Median - A measurement that indicates the center of the data.

Mode - The value that occurs most often.

Mean

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Example:

The accountant of Bill's Sport Shop told Bill, the owner, that the average daily sales for the week were \$150.14. The accountant stressed that \$150.14 was an average and did not represent specific daily sales. Bill wanted to know how the accountant arrived at \$150.14.

Sun.	Mon.	Tues.	Wed.	Thur.	Fri.	Sat.
\$400	\$100	\$68	\$115	\$120	\$68	\$180

$$\text{Mean} = \frac{\$400 + \$100 + \$68 + \$115 + \$120 + \$68 + \$180}{7} = \$150.14$$

Finding the Median of a Group of Values ¹

Example:

Look below at the following yearly salaries of the employees of Rusty's Clothing Shop.

Alice Knight	\$95,000	Jane Wang	\$67,000
Jane Hess	\$27,000	Bill Joy	\$40,000
Joel Floyd	\$32,000		

Find the mean value of all employees.

Solution:

$$\frac{\$95000 + \$27000 + \$32000 + \$67000 + \$40000}{5} = \$52200$$

Weighted Mean

The format for the weighted mean is slightly different from that for the mean. The concept, however, is the same except that you weight each value by how often it occurs (its frequency).

$$\text{Weighted mean} = \frac{\text{Sum of products}}{\text{Sum of frequencies}}$$

Example:

How Jill Rivers calculated her GPA to the nearest tenth given A = 4.0, B = 3.0, C = 2.0, D = 1.0 and F = 0:

Courses	Credit Attempted	Grade Received	Points (Credits × Grade)
Intro to Comp.	4	A	16 (4 × 4)
Psychology	3	B	9 (3 × 3)
English Comp.	3	B	9 (3 × 3)
Business Law	3	C	6 (2 × 3)
Business Math	<u>3</u>	B	<u>9</u> (3 × 3)
	16		49

Solution:

$$\text{Weighted mean} = \frac{\text{Sum of products}}{\text{Sum of frequencies}} = \frac{49}{16} = 3.1$$

Finding the Median of a Group of Values ²

2 Steps

1. Arrange values from the smallest to the largest.  95, 27, 32, 67, 40

2. Find the middle value.  27, 32, 40, 67, 95

Median = 40

- a. Odd number of values: Median is the middle value.
- b. Even number of values: Median is the average of the two middle values.

$$95, 32, 67, 40 \quad \text{Ordered: } 32, 40, 67, 95 \quad \frac{40 + 67}{2} = \mathbf{53.5}$$

Mode

Mode is the value that occurs most often.

If two or more numbers appear most often, you may have two or more modes.

If all the values are different, there is no mode.

Example:

Find the mode of the following data:

3, 4, 5, 6, 3, 8, 9, 3, 5, 3

Solution: **3** is the mode since it is listed 4 times.

Measure of Dispersion

Measure of Dispersion – A number that describes how the numbers of a set of data are spread out or dispersed. **Range and standard deviation are measures of dispersion.**

Range – The difference between the two extreme values (highest and lowest) in a group of values or a set of data.

$$\text{Range} = \text{Highest value} - \text{Lowest value}$$

Example:

Find the range of the following values:

83.6, 77.3, 69.2, 93.1, 85.4, 71.6

$$\text{Range} = 93.1 - 69.2 = \mathbf{23.9}$$

Variance and Standard Deviation

Standard deviation is intended to measure the spread of data around the mean.

6 Steps

1. Find the mean of the set of data.
2. Subtract the mean from each data point to find each deviation.
3. Square each deviation (multiply the deviation by itself).
4. Sum all squared deviations (add them together).
5. Divide the sum of the squared deviations by $n - 1$, where n equals the number of data points. This is the variance.
6. To calculate the standard deviation, find the square root of the number obtained in Step 5 (Use calculator). This is the standard deviation. (the square root is a number that when multiplied by itself equals the amount shown inside the square root symbol).

Calculating the Variance and Standard Deviation

Sample variance

$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

Sample standard deviation

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

Standard Deviation ₁

Data set A	Data set B
$ \begin{array}{cccccccccccccccc} & \overset{x}{1} & \overset{x}{2} & & & \overset{x}{5} & & & & \overset{x}{10} & & \overset{x}{12} & \\ 0 & & & 3 & 4 & & 6 & 7 & 8 & 9 & & 11 & & 13 \\ \hline \text{Mean} & = (1 + 2 + 5 + 10 + 12) \div 5 = 6 \end{array} $	$ \begin{array}{cccccccccccccccc} & & & & \overset{x}{4} & \overset{x}{5} & & & \overset{x}{8} & \overset{x}{9} & & & & \\ 0 & 1 & 2 & 3 & & & 6 & 7 & & & 10 & 11 & 12 & 13 \\ \hline \text{Mean} & = (4 + 4 + 5 + 8 + 9) \div 5 = 6 \end{array} $

Standard Deviation ²

Data sets A and B have equal means, but different standard deviations.

Data	Step 2 Data – Mean	Step 3 (Data – Mean)²
1	1 - 6 = - 5	25
2	2 - 6 = - 4	16
5	5 - 6 = - 1	1
10	10 - 6 = 4	16
12	12 - 6 = 6	36
	Total 0	94(Step 4)

Step 5: Divide by $n - 1$: $\frac{94}{5 - 1} = \frac{94}{4} = 23.5$

Step 6: The square root of $\sqrt{23.5}$ is 4.8 (rounded).

The standard deviation of data set A is 4.8.

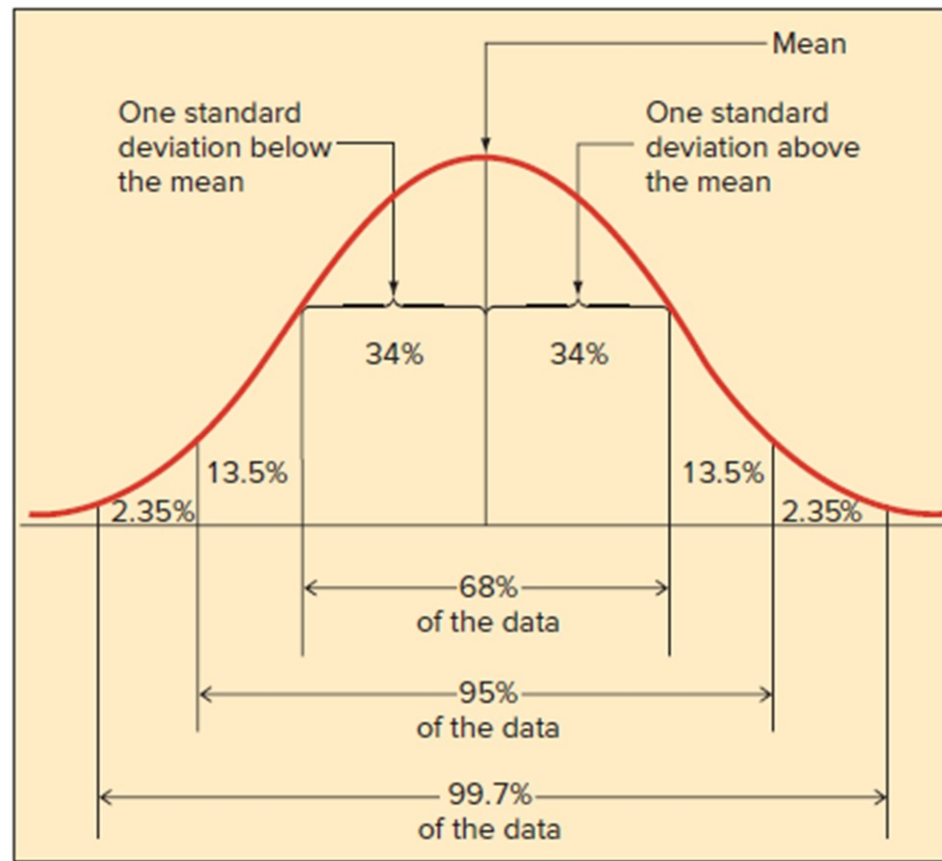
Data	Step 2 Data – Mean	Step 3 (Data – Mean)²
4	4 - 6 = - 2	4
4	4 - 6 = - 2	4
5	5 - 6 = - 1	1
8	8 - 6 = 2	4
9	9 - 6 = 3	9
	Total 0	22(Step 4)

Step 5: Divide by $n - 1$: $\frac{22}{5 - 1} = \frac{22}{4} = 5.5$

Step 6: The square root of $\sqrt{5.5}$ is 2.3

The standard deviation of data set B is 2.3.

Normal Distribution (Figure 5.1)



[Access the text alternative for these images](#)

Normal Distribution (2)

In a normal distribution, the mean, median, and mode are all equal.

Additionally, when the data are normally distributed, the Empirical Rule (Three Sigma Rule) applies, stating that for a normal distribution, also known as a bell curve, approximately:

- 68% of the observations will fall within ± 1 standard deviation of the mean,
- 95% will fall within ± 2 standard deviations of the mean, and
- 99.7% will fall within ± 3 standard deviations of the mean.

Applying the Normal Distribution

5 Steps

1. Calculate the mean.
2. Calculate the standard deviation.
3. Determine the number of standard deviations of the mean.
4. Add and subtract the number of standard deviations to and from the mean.
5. The resulting two values are the range of your data.

Textbook Problem 5-1

Problem Statement:

Calculate the mean (to the nearest hundredth): *LU 5-2(1)*

7, 8, 6, 10

Solution:

$$\frac{7 + 8 + 6 + 10}{4} = \frac{35}{4} = 7.75$$

Textbook Problem 5-8

Problem Statement:

Find the median: $LU\ 5-2(3)$

55, 10, 19, 38, 100, 25

Solution:

$$10, 19, 25, 38, 55, 100 \quad \frac{25 + 38}{2} = \mathbf{31.5}$$

Textbook Problem 5-11

Problem Statement:

Find the mode: $LU\ 5-2(4)$

22, 19, 15, 16, 18, 18, 5, 18

Solution:

= **18**

Textbook Problem 5-12

Problem Statement:

Given:

Truck cost	2023	\$60,000
Truck cost	2008	\$21,000

Calculate the price relative (rounded to the nearest tenth percent):
LU 5-1(3)

Solution:

$$\frac{\$60,000}{\$21,000} \times 100 = \mathbf{285.71\%}$$

Textbook Problem 5-16

Problem Statement:

How many degrees on a pie chart would each be given from the following? *LU 5-1(2)*

Wear digital watch	42%
Wear traditional watch	51%
Wear no watch	7%

Solution:

$$\begin{array}{rcl} .42 \times 360^\circ & = & \mathbf{151.2^\circ} \\ .51 \times 360^\circ & = & \mathbf{183.6^\circ} \\ .07 \times 360^\circ & = & \mathbf{25.2^\circ} \\ \hline & & \mathbf{360^\circ} \end{array}$$

Textbook Problem 5-18

Problem Statement:

Calculate the variance and standard deviation for the following sample set of data.

Round the final answer to the nearest tenth. *LU 5 – 3(2)*

83.6, 92.3, 56.5, 43.8, 77.1, 66.7

Solution:

Data	Step 2 Data – Mean	Step 3 (Data – Mean)²
83.6	83.6 - 70 = 13.6	184.96
92.3	92.3 - 70 = 22.3	497.29
56.5	56.5 - 70 = - 13.5	182.25
43.8	43.8 - 70 = - 26.2	686.44
77.1	77.1 - 70 = 7.1	50.41
66.7	66.7 - 70 = - 3.3	10.89
	Total	1.612.24

$$\text{Variance: } \frac{1612.24}{(6 - 1)} = 322.448$$

Standard deviation:

$$\sqrt{322.448} = 17.69 = 18.0$$

End of Main Content



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